5(a)	Mean power $\langle P \rangle = I_{r.m.s.}^2 R$ $= \left(\frac{2.7}{\sqrt{2}}\right)^2 (50)$ $= 182\text{ W} \quad = 180\text{ W}$	[1] sub [1] ans
5(b)	r.m.s voltage across secondary coil $V_S = I_{r.m.s.} R$ $= \left(\frac{2.7}{\sqrt{2}}\right)(50)$ $= 95.5\text{ V}$	[1] value

	Turns ratio, $\frac{N_P}{N_S} = \frac{V_P}{V_S}$ $\frac{25}{N_S} = \frac{20}{95.46}$ $N_S = \frac{(25)(95.5)}{20}$ $= 119 = 120$	[1] ans
5(c)	For the current in the secondary coil, Period $T = \frac{50 \times 10^{-3}}{3} = 1.67 \times 10^{-2}$ Frequency $f = \frac{1}{1.67 \times 10^{-2}}$ $= 60$ Hz Hence, the frequency of the alternating voltage supply is 60 Hz since it is the same as the frequency of the current through the secondary coil.	[1] correct T [1] ans and statement
5(d)	The values of the root-mean-square current and voltage of the alternating voltage supply are independent of its frequency. The mean power due to the alternating voltage supply is constant. Since the transformer is ideal, the mean power dissipated across R remains unchanged.	[1] [1]
6(a)	Threshold frequency refers to the minimum frequency of the	[1]